Activity 2C: Making a Trumpet with Bling

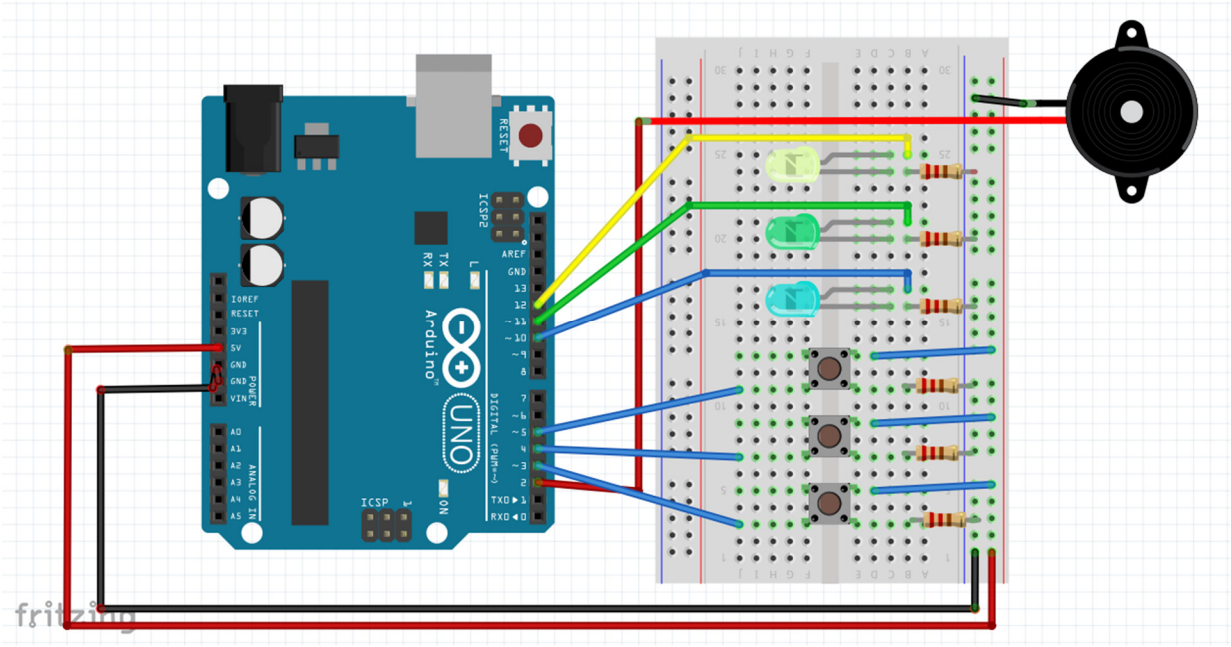
Purpose: To add and modify code the Arduino to use three buttons as inputs to control the outputs which will be both sound frequencies and light.

Materials: Computer, Arduino Uno Board, USB cable, Breadboard, 6 - 220 Ω Resistors, 3 buttons, 10 Jumper Wires, and a potentiometer

New Components:
There are no new components for this activity.

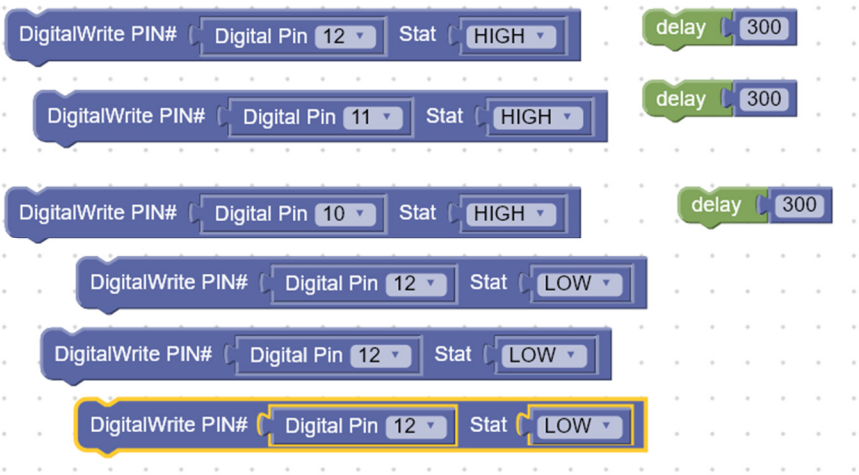
Procedure:

1. Wire the project using the following schematic.



2. You can create your code in the BlocklyDuino Editor then paste it into the Codebender website. Click BlocklyDuino Reference for a link to all the blockly codes.

2. Below are the blocks that you will need to add to your code from Lesson 2B, making music. Assign a working LED to each button so that they go on and off as buttons are pressed. Please look at the schematic so that you assign the correct pins to the lights. If you need help with your code, please watch the following video: **Trumpet With Bling**



*****SAVE your sketch to your BlocklyDuino Folder that you created*****

6. Copy the Arduino Code from BlocklyDuino into the Codebender website. Hook up your arduino to your computer using the USB cable and click the arrow button to upload your code to the Arduino microcontroller.

7. Your program is now loaded. Observe your LED's and pay attention to the tones when you play your buttons and complete your worksheet titled "Assignment 2C: Making a Trumpet with Bling."

Troubleshooting: If your circuit is not working!

1. Double check your code! Did you put in every block? Are the settings correct?
2. Refresh the website and reload the code.
3. The next step is to check your circuit!
    a. Check your wiring! Is it the same as the circuit diagram? Does the circuit go from the positive numbered pin to the ground?
    b. Next check each component. Check the leds, is the long positive arm connected

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by wires to the pins?

c. Next inserting the LEDs directly into pin 13 and the adjacent ground. Does it light when the your code is on?

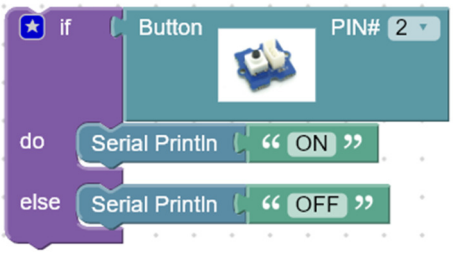


d. Check the speaker. Rewire it to pin A0. Open a new Blocklyduino code and test the speaker by playing this code:



This should cause the speaker to chirp.

e. Check the buttons one at a time. This can be achieved by connecting it and using the serial monitor built into Codebender. Open a new window of Blocklyduino and load this test program. Change the pin number to each of the button pins and test one at a time.



Open the serial monitor in Codebender and select “connect”As you hold the button down the serial monitor should change to on from off.

f. Next check the individual wires in the circuit. This can be done by connecting an LED to pin 13 only then using the wire to ground the short pin. Do not connect the pin to the ground directly. (Shorting the wires) This will damage the UNO.

g. For further information on using the serial monitor see Activity 4A Using a Serial Monitor Display.

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Assignment 2C WORKSHEET: Making a Trumpet with BLING

1. What happens when you press the buttons?

Extension Challenges:

2. Change the delay of the light to 500 milliseconds and then change the duration of the tone to match. Load you program and describe how the program changed.

3. Add a fourth note and attach an RGB LED to the fourth note. Refer to lesson 1H on how to wire and code a RGB LED.

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